

COURSE PROJECT REPORT
ON
**Investigating the Effect of Training Data Size on
Machine Learning Performance Using the Wine Quality Dataset**

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1. Research Question

Does more training data always improve ML performance?

2. Hypothesis

Increasing the amount of training data will generally improve model performance, but the improvement will not necessarily continue at every training size; performance may level off or fluctuate as additional data is added.

3. Theoretical Background

Machine learning is the process of learning patterns from data so that a model can make predictions on previously unseen observations. This investigation uses supervised learning, where the model learns a relationship between input features and a known target variable. The main question is not simply whether a model can predict wine quality, but how its performance changes when the amount of training data is varied.

Linear Regression models a numerical target as a linear combination of input features. In this investigation, the model is trained separately using five progressively larger subsets of the available training pool. The same fixed test set is used for every condition so that differences in the reported metrics are associated with the training-data condition as consistently as possible.

Mean Absolute Error (MAE) measures the average absolute difference between predicted and actual values; lower values indicate smaller average errors. Root Mean Squared Error (RMSE) also measures prediction error, but gives greater influence to larger errors; lower values are better. R^2 indicates the proportion of variation in the target explained by the model; higher values indicate better explanatory performance.

Training-data size can affect the stability of learned parameters. With fewer observations, the model may be more sensitive to the particular samples included. Additional data can therefore improve the representation of the underlying relationship. However, more observations do not guarantee that every added sample will improve performance. Bias refers to systematic error associated with model assumptions, while variance refers to sensitivity to the particular training sample. Diminishing returns describes a situation in which the benefit from additional data becomes smaller after sufficient data is already available.

In this investigation, bias and variance are discussed as theoretical concepts rather than directly measured quantities. The experiment is designed to observe the empirical relationship between training-data size and test-set performance.

4. Dataset / Data Source

The investigation uses the Wine Quality Dataset, specifically the red wine quality data used in the experiment. The original dataset contained 1,599 samples. Duplicate rows were removed during preprocessing, resulting in 1,359 cleaned samples.

The target variable is wine quality. The cleaned target distribution is uneven: quality 3 contains 10 observations, quality 4 contains 53, quality 5 contains 577, quality 6 contains 535, quality 7 contains 167, and quality 8 contains 17. Thus, quality levels 5 and 6 account for most observations in the dataset.

5. Experimental Design

An 80/20 train-test split was used after preprocessing. This produced a training pool of 1,087 samples and a fixed test set of 272 samples. The same fixed test set was used for all five training-data conditions. This provides a consistent basis for comparing model performance across different training sizes.

The five training conditions were 20%, 40%, 60%, 80%, and 100% of the 1,087-sample training pool, corresponding to 217, 434, 652, 869, and 1,087 training samples respectively. A separate Linear Regression model was trained for each condition.

StandardScaler was used in preprocessing. For each training condition, the scaler was fitted using only the corresponding training subset and was then applied to the fixed test set. The model was evaluated using MAE, RMSE, and R^2 .

The experimental design keeps the evaluation set fixed while changing the amount of training data. Therefore, the investigation focuses on how the quantity of training data is associated with test-set performance under the chosen dataset, model, preprocessing procedure, and subset-selection method.

Training Conditions

Training Data	Training Samples
20%	217
40%	434
60%	652
80%	869
100%	1,087

Table 1: Training-data conditions used in the investigation

6. Results

The final experimental results for the five training-data conditions are presented below. Lower MAE and RMSE values indicate lower prediction error, while a higher R^2 indicates greater explanatory performance.

Training Data	Training Samples	MAE	RMSE	R^2
20%	217	0.510564	0.674985	0.356814
40%	434	0.516034	0.674005	0.358679
60%	652	0.503097	0.656459	0.391636
80%	869	0.502285	0.651987	0.399896
100%	1,087	0.504141	0.656513	0.391536

Table 2: Model Performance at Different Training Data Sizes

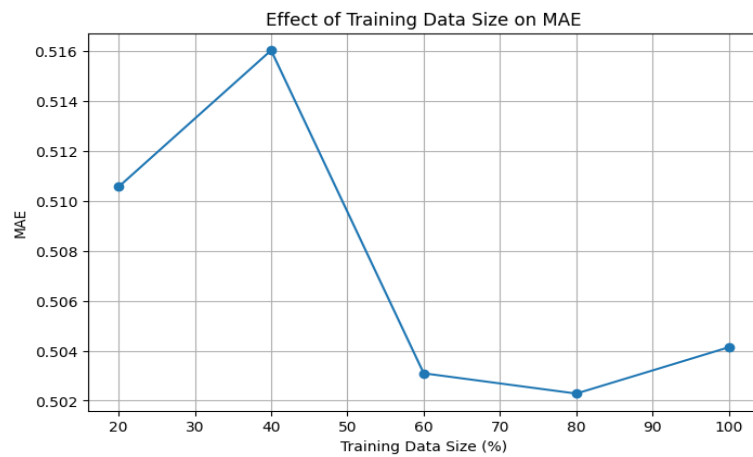


Figure 1: Effect of Training Data Size on MAE

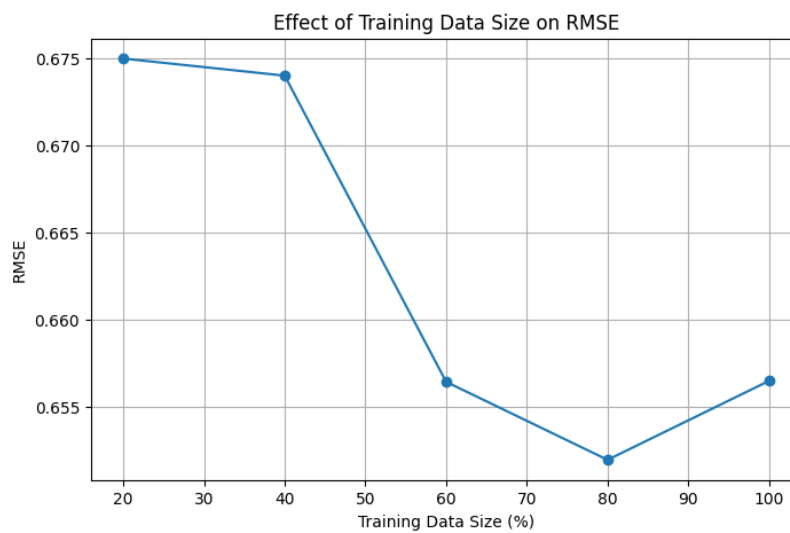


Figure 2: Effect of Training Data Size on RMSE

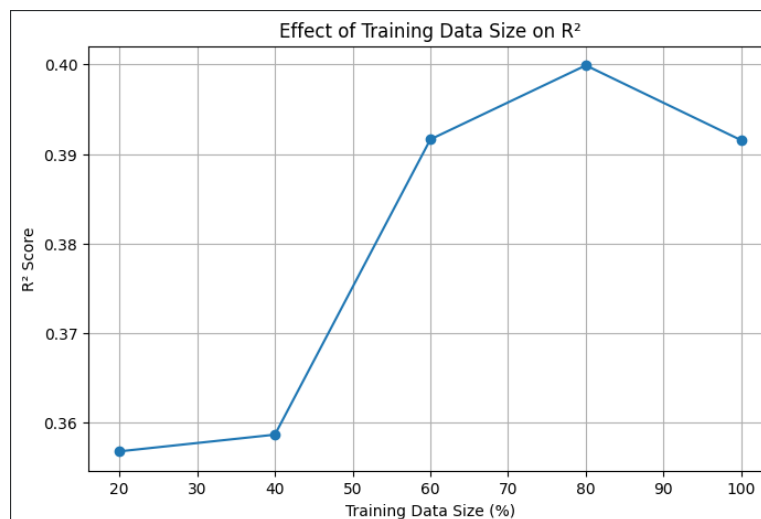


Figure 3: Effect of Training Data Size on R^2

7. Results Interpretation

The R^2 value increased from 0.356814 at 20% training data to 0.399896 at 80%. The 100% condition then decreased slightly to 0.391536. RMSE followed a similar overall pattern: it decreased from 0.674985 at 20% to 0.651987 at 80%, then increased slightly to 0.656513 at 100%. MAE fluctuated rather than improving continuously: it increased from 0.510564 to 0.516034 between 20% and 40%, then decreased through 80%, before increasing slightly to 0.504141 at 100%.

The clearest improvement occurs as the experiment moves from smaller training sets toward the 60% and 80% conditions. The highest R^2 and lowest RMSE are both observed at 80% in this particular experiment. At 100%, all three metrics are slightly worse than at 80%. Therefore, the results do not show continuous improvement at every training size.

8. Analysis and Interpretation

Why performance initially improved

Smaller training sets provide fewer observations from which Linear Regression can estimate the relationship between the physicochemical features and wine quality. Increasing the number of training observations can provide more information for estimating model coefficients and may make the fitted relationship more representative of the available data. This provides a reasonable theoretical explanation for the improvement observed from the smaller conditions toward 60% and 80%.

Why the improvement was not perfectly smooth

Adding observations does not guarantee that every newly included sample will improve test performance. The composition of the selected subset, noise in observations, the uneven target distribution, the amount of new information contributed by the added samples, the assumptions of Linear Regression, and the ordering used to construct progressively larger subsets may all influence the observed values. These are possible explanations rather than causes proven by this experiment.

Diminishing returns

The results are consistent with a leveling-off effect: the benefit of additional training data becomes smaller after the model has already received a substantial amount of data. However, the experiment does not establish a perfectly smooth diminishing-return curve because the MAE fluctuates and the 100% condition is slightly worse than the 80% condition.

Bias and variance

Linear Regression has assumptions about the form of the relationship between input variables and the target. Increasing the amount of data cannot remove limitations caused by an unsuitable model form. R^2 remains below 0.40 in every condition, so a substantial amount of target variation remains unexplained by the model, although the experiment does not establish the exact cause. Smaller training sets can also make parameter estimates more sensitive to sample composition, which is related to variance. Variance was not directly measured because each training size was not repeated using multiple independently sampled training sets.

9. Data Quality Analysis

The cleaned dataset contains an uneven distribution of wine-quality values. Quality 5 and quality 6 contain 577 and 535 observations respectively, while quality 3 and quality 8 contain only 10 and 17 observations. This means that adding training samples does not necessarily add equal representation across all target values.

Data quantity and data quality are different factors. The investigation changes training-data quantity while keeping the dataset and preprocessing procedure fixed; it does not independently vary the quality, diversity, or composition of the data. Therefore, the target distribution is an important contextual factor when interpreting the results, but the experiment does not establish a direct causal effect of target imbalance on the observed performance changes.

10. Critical Evaluation and Limitations

- Single dataset: only the red Wine Quality Dataset was used, so the findings cannot automatically be generalized to other datasets.
- Single model: only Linear Regression was evaluated. Other machine-learning models may respond differently when training-data size changes.
- Fixed train-test split: only one fixed train-test split was used, and a different split could produce somewhat different results.
- Training subset selection: progressively larger subsets were selected according to dataset ordering rather than repeated random sampling, so exact results may partly depend on which samples were included at each size.
- Data quality: data quality was not independently varied in the experiment.
- Bias and variance: these concepts were discussed theoretically but were not directly measured.
- Small differences: the difference between the 80% and 100% conditions is relatively small, so the observed decrease should not be interpreted as strong evidence that additional data is harmful.

11. Conclusion

This investigation examined whether more training data always improves machine-learning performance. For the Wine Quality Dataset and Linear Regression model used here, performance generally improved as the training set increased from 20% to 80%. The 80% condition produced the highest R^2 and the lowest RMSE, while MAE was also lowest at 80%. When the full 100% training pool was used, MAE, RMSE, and R^2 were all slightly worse than at 80%.

Therefore, more training data did not guarantee continuous improvement in this experiment. The result is specific to the selected dataset, model, preprocessing procedure, fixed test set, and training-subset selection method. Additional data can be useful, but its benefit depends on factors such as sample composition, data quality, model characteristics, and the information contained in the added observations.

12. AI Usage Declaration

AI tools were used as supporting tools during the development of this investigation. They assisted with understanding machine-learning concepts, planning the investigation, generating and debugging Python code, explaining methodology, interpreting results, preparing visualizations, and organizing the report. The experiment was executed by the students in Google Colab, and the numerical results presented in this report were obtained from the students' own experiment. AI-generated suggestions were reviewed and verified against the actual experiment before inclusion. Where an AI suggestion conflicted with the final experimental output, the suggestion was corrected or rejected and the students' verified results were retained.

References

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